SEAGAS System Guide: What We Built and How To Use It

Group 10 | COS40005 Computing Technology Project A | POC004 | 1 November 2026
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In short: SEAGAS compares a student's skills with what a job asks for, gives a **Job Readiness Score**, shows each skill as Strong, Developing or a Gap with the reason, and recommends learning resources. Advisors see how their students are doing; admins manage accounts and settings. This guide lists what we built, then gives the steps for **each role** from the first start to the end of the process. Scores support decisions; they do not predict hiring.

1. What We Built

1.1 The system in numbers

Item	Now in the system
Roles	Student, Advisor (Dean or Head of Programme), Admin
Courses	Computer Science, Information and Communications Technology, Business (13 programmes in total)
Library	166 skills, 10 job roles, 363 learning resources (active entries in the local database, 1 November)
Test data	5,000 synthetic students, all assessed; 350 job descriptions; no real student data
Skill extraction	Keyword, semantic and hybrid matching. On the 60-JD gold set, hybrid scored best (micro F1 0.731)
Automated tests	185 backend tests pass. NLP tests: 355 pass and 1 fails (a non-IT sample the test should leave out; the website is not affected)

1.2 Main functions

Area	What it does
Accounts and security	No self-registration; advisors submit new students and admins create the accounts. 9-digit student IDs and 5-digit staff IDs, temporary passwords that must be changed at first sign-in, role-based access, and an audit log. Login details are emailed.
Student profile	Academic details with GPA and CGPA (reminder each semester), skills grouped by category, work experience, projects and certifications with evidence, CV upload with skill extraction, and Export CV.
Assessment	A role template or a pasted job description; Job Readiness Score, seven dimensions, and a stored reason for every skill. Missing information shows "No data yet" rather than a misleading 0%.
Development	Explainable recommendations (Free, Free to learn, Paid), activities that need evidence before completion, progress over time, Run again, and Career Paths.
Advisor portal	Cohort charts, skill gaps by programme, job role and skill category, sortable student list, learning progress, CSV export, student pages with CV and printable report, and notes that agree or disagree with AI results. Advisors manage the library entries of their own courses.

Admin portal	User Management, Student Intake with Email outbox, Skills & Roles, Learning Resources, Audit Log and System Settings.

1.3 Who worked on what

Member	Main work
Yan Min Xuan (Shanice)	Team leader; coordination, student-facing UI, Figma prototype, report and documentation.
Tan Jun Xiong (Aaron)	Co-leader; recommendation library and rule-based engine, coordination and documentation.
Soh Way Miin (Carl)	Backend, database and security; NLP integration; accounts and intake; advisor and admin functions; email.
Tee Ren Hang	React frontend, UI/UX, and the first admin, advisor and student pages.
Ng Yong Hin	NLP extraction and matching, job-description dataset, gold set and method comparison, Business skills and synthetic data.

2. Starting the System

You need: Windows, Python 3.10+, Node.js 18+ and PostgreSQL 17 (user `postgres`, password `postgres123`).

1. Get the latest code: `git pull` on the `dev` branch.
2. Double-click **start_all.bat** in the project folder. Two windows open: backend and frontend. Run this again after every pull; it updates the database safely.
3. **First start only:** the backend window asks you to create the first admin. The email must end with `@admin.newinti.edu.my`.
4. Open **http://localhost:5173** and click to sign in.

Sending real email (optional): add the `SMTP_` settings to `backend/.env` on that computer and restart. Without them the system still works, and emails stay in **Student Intake → Email outbox**. Never commit this file or share its password. See `SEAGAS_TeamUpdate_1November_By_Waymiin.pdf`, section 3.

> **Accounts for quick testing.** Synthetic student: `student0001@synthetic.example.com`, student ID `900000001`, password `Seagas@2026`. Advisors and admins sign in with the accounts created on your own computer.

3. The Whole Process at a Glance

1. Admin Creates advisor accounts	**2. Advisor** Uploads the new-student list	**3. Admin** Assigns IDs, creates student accounts	**4. Student** Signs in, sets a new password
5. Student Completes profile and skills	**6. Student** Runs an assessment, reads the gaps	**7. Student** Does activities, adds evidence, runs again	**8. Advisor** Reviews progress, adds notes

4. Admin: Step by Step

4.1 Create the advisor accounts

1. Sign in with your email and password. Leave **Student ID** empty.
2. Open **User Management** and click **+ Add user**.
3. Choose role **Advisor**, then enter the full name and personal email.
4. Choose the **Position**. A free staff ID is suggested, and the school email follows from it. Each office has only one active Dean.
5. Click **Create account**. The advisor receives the login details by email.

Position	Courses the advisor sees	Staff ID starts with
Dean, Computing	Computer Science and ICT	10
Head of Programme, Computer Science	Computer Science	11
Head of Programme, ICT	ICT	12
Dean, Business	Business	20
Head of Programme, Business	Business	21

4.2 Create the student accounts

1. After an advisor uploads a list (section 5.1), open **Student Intake** → **New student lists** and open the list.
2. Click **Suggest student IDs**. Each student gets a student ID and school email; you can edit them.
3. Click **Reject** for anyone who should not get an account.
4. Click **Create N accounts**. Each student receives their login details by email.
5. If no mail server is set up, open the **Email outbox** tab, read each email and pass the details to the student.

4.3 Ongoing work

- **User Management:** search accounts, **Edit** names, school emails, programmes, positions and staff IDs, **Reset password**, and **Deactivate** accounts.
- **Skills & Roles / Learning Resources:** add entries, edit those added on the website, and switch any entry on or off. Entries from the seed files can only be switched on or off.
- **System Settings:** semester start dates, readiness bands (defaults 70 and 40) and upload limits. Check the mail server and send a test email here.
- **Email outbox:** check delivery and resend. Look at the recipient first.
- **Audit Log:** see sign-ins, assessments, advisor views and admin changes.

5. Advisor: Step by Step

5.1 First sign-in and new students

1. Sign in with the school email and temporary password from your email. Leave **Student ID** empty.
2. Choose your own password: at least 8 characters, with an uppercase letter, a lowercase letter, a digit and a symbol.
3. Open **New Students** and click **Download the template**.
4. Fill in Full Name, Phone Number, Personal Email, Course and Programme. Course and Programme are drop-down lists.
5. Choose the file and click **Upload**. An admin then creates the accounts (section 4.2); follow the status under **My uploaded lists**.

5.2 Follow your students (Cohort Overview)

1. If you look after two courses, choose one at the top or stay on **All my courses**.
2. **Overview:** read the readiness distribution, student status, top skill gaps and target roles. Click a status to list those students, including those with no assessment.
3. **Skill Gaps:** filter by programme and assessed job role. Click a category card, such as **Soft skills**, to see only that category.
4. **Students:** search, filter and sort, then click a name to open the student's page.
5. **Learning Progress:** see the most completed resources and each student's activity.
6. Use **Export CSV** to take a table out of the system.

5.3 Review one student

1. Read the profile, latest assessment, skills against requirements, readiness over time, evidence and activities.
2. Download the student's CV if there is one, or click **Print report**.
3. Under **Advisor notes**, add a general note, or agree or disagree with the latest AI result and give a reason. Students never see these notes.
4. Use your own judgement: the score is decision support, not a final decision.

5.4 Manage the library of your courses

1. Open **Skills & Roles** to add skills and job roles for your courses. After adding a role, use **Skills** to choose what it requires.
2. Open **Learning Resources** and click **+ Add resource** for a skill your students see.
3. Entries shown as **Shared · admin only** are also used by other courses. Ask an admin to change them.

6. Student: Step by Step

6.1 First sign-in

1. Find the email with your student ID, school email and temporary password. Check the spam folder too.
2. Sign in with all three. Students must enter the **Student ID**.
3. Choose your own password. The temporary one stops working after that.

6.2 Build your profile (My Profile)

1. **Info:** enter university, GPA, CGPA, expected graduation, year of study and target role, then click **Save Profile**. Your course and programme are set by the university.
2. **Skills:** choose a skill and level and click **Add**. Or upload a PDF or Word CV, tick the skills found, and confirm.
3. **Experience:** add internships, jobs, freelance work or volunteering.
4. **Projects** and **Certifications:** add them with evidence (a file or a link; for videos, use a link). A project or certification can make a beginner skill count as Strong.

6.3 Run an assessment (Job Analysis)

1. Choose **Use Role Template** and select a role, or choose **Paste Job Description** and paste the text.
2. Keep the matching method on **Hybrid** (recommended).
3. Click **Run Assessment**. You need at least one skill in your profile.
4. Read your Job Readiness Score, then click **View Skill Gap**.

6.4 Understand the result (Skill Gap)

You see	It means
Strong	You have the skill at intermediate or advanced level, or at beginner level with a project, certification or internship.
Developing	A beginner skill without that evidence, or a different skill that is similar or related.
Gap	Nothing in your profile is close enough.
Not required	This job asks for no skill of that category.
No data yet	Nothing is recorded in your profile for it. Add it and run the assessment again.

Hover over a skill to read why it is in that group.

6.5 Improve and track (Recommendations, Progress)

1. Open **Recommendations** and choose **Free**, **Free to learn** or **Paid**.
2. Click **Start** on an activity and open its link to do the learning.
3. When finished, click **Mark Complete** and attach evidence (a file or a link). Completion needs evidence.
4. Add the new skill on **My Profile** when the system offers.
5. Open **Progress** and click **Run again**. Your new score appears with the change since last time.

6.6 Plan your career and share your record

- **Career Paths:** see the job roles of your course ranked by your skills, with what is missing. Click **Assess me against this role** or **Make this my target role**.
- **Export CV** (My Profile): click **Save as PDF** and choose "Save as PDF" as the printer.
- **Each semester** (January, May, September): when the reminder appears, update your GPA and CGPA or confirm there is no change.

7. A Full Demo From Start to End

Use an email address you can open for the new student. With a mail server set up, the system really sends to it.

Step	Role	What to do and show
1	Admin	Create a Head of Programme, Computer Science (+ Add user).
2	Advisor	Sign in, change the password, download the template, and upload one new student.
3	Admin	Student Intake: Suggest student IDs, then Create 1 account. Show the email.
4	Student	Sign in, change the password, complete Info, and add skills by hand and from a CV.
5	Student	Job Analysis: run an assessment. Show the score, reasons and "No data yet".
6	Student	Recommendations: Start an activity, Mark Complete with evidence, and add the skill.
7	Student	Progress: Run again and show the score change. Show Career Paths and Export CV.
8	Advisor	Find the student, view the evidence, add a note that agrees or disagrees with the AI result, and print the report.
9	Advisor	Skill Gaps: filter by programme and open Soft skills, using the 5,000 synthetic students.
10	Admin	Audit Log: show that the steps above were recorded.

Before a demo: the full process above has not yet been run from start to end in a browser. Please rehearse it once. Sign-in links in emails point to `localhost`, so they open only on the computer running the system unless `FRONTEND_URL` is set in `backend/.env`.

8. Common Problems

Problem	What to do
Student cannot sign in	Students must enter the school email, the 9-digit student ID and the password. Staff leave the student ID empty.
Forgotten password	An admin uses Reset password in User Management. The user then chooses a new one.
No email arrived	Check the spam folder, then Student Intake → Email outbox. An admin can read or resend it.
"Add at least one skill"	Add a skill on My Profile → Skills, then run the assessment again.
A page cannot load after a pull	Close both windows and run start_all.bat again to update the database.
Hybrid shows a notice	The optional NLP packages are not installed, so keyword matching is used. Install `nlp/requirements.txt` and restart.